COMPUTER ENGINEERING
QUEZON CITY

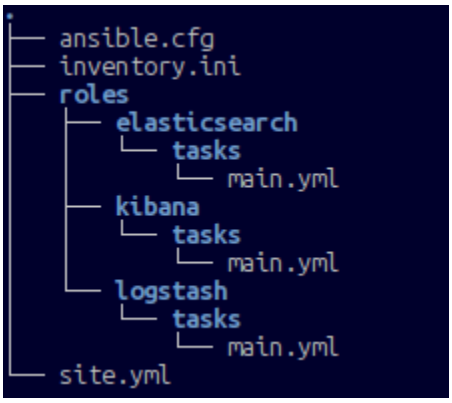
COURSE CODE/TITLE:	CPE 21 - Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	2026-10-01	**DATE SUBMITTED:**	2026-10-06
NAME:	Ranzel Aldous Gabriel L. Duenas	**INSTRUCTOR:**	Engr. Jimlord Quejado

ACTIVITY NO. 10.1 **Install, Configure and Manage Enterprise Log Monitoring via Ansible**

1. PROCEDURE OUTPUT

1. **Create a playbook that:**
   a. Install and configure Elastic Stack in separate hosts (Elastic Search, Kibana, Logstash)

2. **Apply the concept of creating roles.**
   a. **Structural tasks**



- Defines the project tree structure and configure inventory.ini mapping all three hosts (192.168.56.104, 192.168.56.105, and 192.168.56.108) so each target node executes all three Ansible roles.

   b. **inventory.ini**

```
[ubuntu_servers]
192.168.56.104 ansible_user=ubuntu
192.168.56.105 ansible_user=ubuntu

[centos_servers]
192.168.56.108 ansible_user=centos

[all_servers:children]
ubuntu_servers
centos_servers
```

   c. **elasticsearch**/tasks/main.yml

```
---
# Debian / Ubuntu Tasks
- name: Clean APT cache to free disk space (Debian/Ubuntu)
  ansible.builtin.apt:
    clean: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install prerequisites (Debian/Ubuntu)
  ansible.builtin.apt:
    name:
      - apt-transport-https
      - openjdk-11-jdk
      - gpg
      - curl
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

- name: Create Keyrings directory (Ubuntu)
  ansible.builtin.file:
    path: /etc/apt/keyrings
    state: directory
    mode: '0755'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic GPG key (Ubuntu)
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /etc/apt/keyrings/elasticsearch.asc
    mode: '0644'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic APT repository (Debian/Ubuntu)
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/etc/apt/keyrings/elasticsearch.asc]
https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
  when: ansible_facts['os_family'] == "Debian"

- name: Install Elasticsearch (Debian/Ubuntu)
  ansible.builtin.apt:
    name: elasticsearch
    state: present
```

```
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

# RedHat / CentOS Tasks
- name: Import Elastic GPG Key (RedHat/CentOS)
  ansible.builtin.rpm_key:
    state: present
    key: https://artifacts.elastic.co/GPG-KEY-elasticsearch
  when: ansible_facts['os_family'] == "RedHat"

- name: Add Elastic YUM Repository (RedHat/CentOS)
  ansible.builtin.yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"

- name: Install Elasticsearch (RedHat/CentOS)
  ansible.builtin.yum:
    name: elasticsearch
    state: present
  when: ansible_facts['os_family'] == "RedHat"

# Common Configurations
- name: Configure Elasticsearch network binding
  ansible.builtin.lineinfile:
    path: /etc/elasticsearch/elasticsearch.yml
    regexp: '^#?network.host:'
    line: 'network.host: 0.0.0.0'

- name: Start and enable Elasticsearch service
  ansible.builtin.systemd:
    name: elasticsearch
    state: started
    enabled: yes
```

d. **kibana**/tasks/main.yml

```
---
# Debian / Ubuntu Tasks
- name: Clean APT cache to free disk space (Debian/Ubuntu)
```

```
  ansible.builtin.apt:
    clean: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Install OpenJDK (Debian/Ubuntu)
  ansible.builtin.apt:
    name: openjdk-11-jdk
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

- name: Create Keyrings directory (Ubuntu)
  ansible.builtin.file:
    path: /etc/apt/keyrings
    state: directory
    mode: '0755'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic GPG key (Ubuntu)
  ansible.builtin.get_url:
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /etc/apt/keyrings/elasticsearch.asc
    mode: '0644'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic APT repository (Debian/Ubuntu)
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/etc/apt/keyrings/elasticsearch.asc]
https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
  when: ansible_facts['os_family'] == "Debian"

- name: Install Logstash (Debian/Ubuntu)
  ansible.builtin.apt:
    name: logstash
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

# RedHat / CentOS Tasks
```

```
- name: Install OpenJDK (RedHat/CentOS)
  ansible.builtin.yum:
    name: java-11-openjdk
    state: present
  when: ansible_facts['os_family'] == "RedHat"

- name: Add Elastic YUM Repository (RedHat/CentOS)
  ansible.builtin.yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"

- name: Install Logstash (RedHat/CentOS)
  ansible.builtin.yum:
    name: logstash
    state: present
  when: ansible_facts['os_family'] == "RedHat"

# Common Configurations
- name: Start and enable Logstash service
  ansible.builtin.systemd:
    name: logstash
    state: started
    enabled: yes
```

e. **logstash**/tasks/main.yml

```
---
# Debian / Ubuntu Tasks
- name: Clean APT cache to free disk space (Debian/Ubuntu)
  ansible.builtin.apt:
    clean: yes
  when: ansible_facts['os_family'] == "Debian"

- name: Create Keyrings directory (Ubuntu)
  ansible.builtin.file:
    path: /etc/apt/keyrings
    state: directory
    mode: '0755'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic GPG key (Ubuntu)
  ansible.builtin.get_url:
```

```
    url: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    dest: /etc/apt/keyrings/elasticsearch.asc
    mode: '0644'
  when: ansible_facts['os_family'] == "Debian"

- name: Add Elastic APT repository (Debian/Ubuntu)
  ansible.builtin.apt_repository:
    repo: "deb [signed-by=/etc/apt/keyrings/elasticsearch.asc]
https://artifacts.elastic.co/packages/8.x/apt stable main"
    state: present
    filename: elastic-8.x
  when: ansible_facts['os_family'] == "Debian"

- name: Install Kibana (Debian/Ubuntu)
  ansible.builtin.apt:
    name: kibana
    state: present
    update_cache: yes
    lock_timeout: 180
  environment:
    DEBIAN_FRONTEND: noninteractive
  when: ansible_facts['os_family'] == "Debian"

# RedHat / CentOS Tasks
- name: Add Elastic YUM Repository (RedHat/CentOS)
  ansible.builtin.yum_repository:
    name: elastic-8.x
    description: Elastic repository for 8.x packages
    baseurl: https://artifacts.elastic.co/packages/8.x/yum
    gpgcheck: yes
    gpgkey: https://artifacts.elastic.co/GPG-KEY-elasticsearch
    enabled: yes
  when: ansible_facts['os_family'] == "RedHat"

- name: Install Kibana (RedHat/CentOS)
  ansible.builtin.yum:
    name: kibana
    state: present
  when: ansible_facts['os_family'] == "RedHat"

# Common Configurations
- name: Configure Kibana server host
  ansible.builtin.lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?server.host:'
    line: 'server.host: "0.0.0.0"'
```

```
- name: Configure Kibana localhost Elasticsearch URL
  ansible.builtin.lineinfile:
    path: /etc/kibana/kibana.yml
    regexp: '^#?elasticsearch.hosts:'
    line: 'elasticsearch.hosts: ["http://localhost:9200"]'

- name: Start and enable Kibana service
  ansible.builtin.systemd:
    name: kibana
    state: started
    enabled: yes
```

f. site.yml

```
---
- name: Deploy Complete Elastic Stack on Ubuntu Servers
  hosts: ubuntu_servers
  become: true
  roles:
    - elasticsearch
    - logstash
    - kibana

- name: Deploy Complete Elastic Stack on CentOS Server
  hosts: centos_servers
  become: true
  roles:
    - elasticsearch
    - logstash
    - kibana
```

3. **Describe how you did step 1. (Provide screenshots and explanations in your report. Make your report detailed such that it will look like a manual.)**
   a. **Debian / Ubuntu Tasks**
      - **Clean APT cache:** Removes stored .deb files from /var/cache/apt/archives to free up system space before installation.
      - **Install prerequisites:** Installs necessary tools (openjdk-11-jdk for Java, gpg for keys, curl for downloads, apt-transport-https for HTTPS repos).
      - **Create Keyrings directory:** Ensures /etc/apt/keyrings exists to store repository security keys securely.
      - **Add Elastic GPG key:** Downloads Elastic's official security signature key to verify package authenticity.
      - **Add Elastic APT repository:** Adds Elastic's official package repository to Ubuntu's software source list.
      - **Install [Elasticsearch / Logstash / Kibana]:** Downloads and installs the requested Elastic Stack software package via apt.
   b. **RedHat / CentOS Tasks**

- **Import Elastic GPG Key:** Adds Elastic's cryptographic security key directly to RPM's key manager.
- **Add Elastic YUM Repository:** Creates a .repo configuration file so yum knows where to download Elastic packages.
- **Install [Elasticsearch / Logstash / Kibana]:** Downloads and installs the package via yum (including Java dependencies like java-11-openjdk).

c. Common Configurations

- **Configure network/server binding:** Modifies configuration files (elasticsearch.yml or kibana.yml) to bind services to 0.0.0.0 so they are accessible across your network.
- **Configure Kibana Elasticsearch URL:** Points Kibana to connect to http://localhost:9200 where Elasticsearch runs.
- **Start and enable service:** Starts the background service immediately and enables it to launch automatically when the server boots.

4. Show an output of the installed Elastic Stack for both Ubuntu and CentOS.

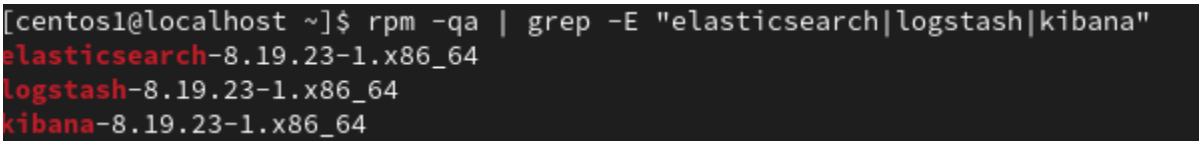
ubuntu1@192.168.56.104



ubuntu2@192.168.56.105



centos



5. Make sure to create a new repository in GitHub for this activity.

GitHub Link: https://github.com/mo-na-min/Enterprise-Log-Monitoring

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What are the benefits of having log monitoring tool?
   - A log monitoring tool gives one place to watch system events as they happen and to see how apps are doing. This helps developers spot operational problems early and deal with them quickly. It also helps find possible security risks.

Conclusion:

We set up an automated way to deploy and run the Elastic Stack. The stack included Elasticsearch, Logstash, and Kibana. We used custom Ansible roles to handle this across different Linux systems. Some hosts ran Debian or Ubuntu. Others ran RedHat or CentOS. We also adjusted the playbook tasks. We worked around system limits during the setup. This included storage partitioning and network binding rules. After those changes, the logging setup stayed stable. It also worked across multiple nodes for log monitoring.

In the end, the work showed why infrastructure-as-code matters. It also showed that planning resources ahead of time helps keep system admin tasks scalable.

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3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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